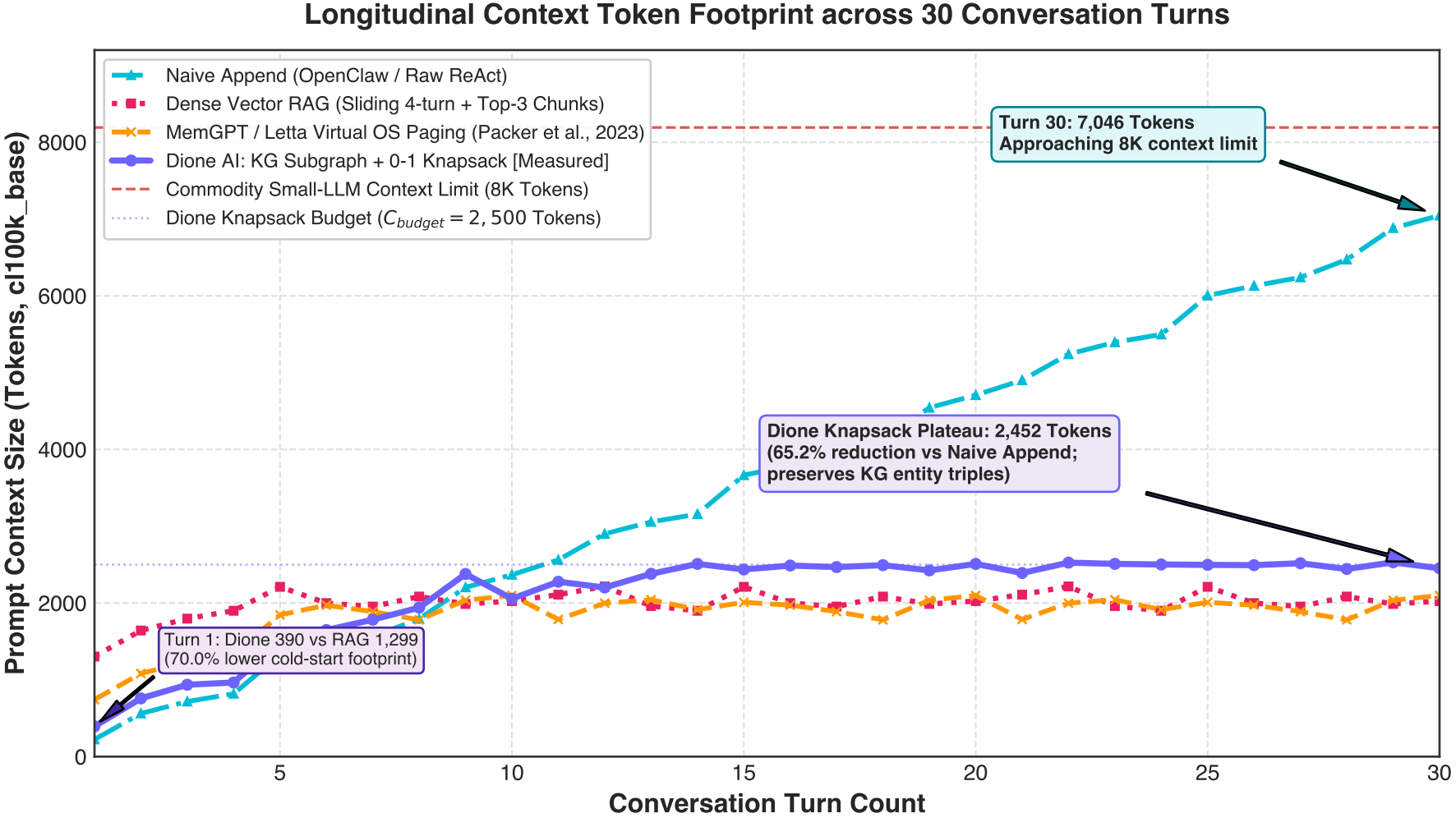
Longitudinal Context Token Footprint across 30 Conversation Turns
Naive Append (OpenClaw / Raw ReAct)
Dense Vector RAG (Sliding 4-turn + Top-3 Chunks)
MemGPT / Letta Virtual OS Paging (Packer et al., 2023)
Dione AI: KG Subgraph + 0-1 Knapsack [Measured]
Commodity Small-LLM Context Limit (8K Tokens)
Dione Knapsack Budget ($C_{budget} = 2,500$ Tokens)
Turn 30: 7,046 Tokens
Approaching 8K context limit
Dione Knapsack Plateau: 2,452 Tokens
(65.2% reduction vs Naive Append;
preserves KG entity triples)
Turn 1: Dione 390 vs RAG 1,299
(70.0% lower cold-start footprint)
Prompt Context Size (Tokens, cl100k_base)
0
2000
4000
6000
8000
5
10
15
20
25
30
Conversation Turn Count